বরেন্দ্র বিশ্ববিদ্যালয়

VARENDRA UNIVERSITY

LAB MANUAL

FOR

Course Title: Computer Architecture Lab
Course Code: CSE 324

Department of
Computer Science and Engineering
Varendra University
Rajshahi, Bangladesh

VARENDRA UNIVERSITY

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বিশ্ববিদ্যালয়

Varendra University

Department of Computer Science & Engineering

CSE 324

Computer Architecture Laboratory

Student ID	
Student Name	
Section	
Name of the Program	
Name of the	

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Lab Instructions:

- Students should come with thorough preparation for the experiment to be conducted.
- Students will not be permitted to attend the laboratory unless they bring the practical record fully completed in all respects pertaining to the experiment conducted in the previous class.
- Be honest in developing and representing your program. If a particular program output appears wrong repeat the program carefully.
- Strictly observe the instructions given by the Faculty / Lab. Instructor.
- Take permission before entering in the lab and keep your belongings in the racks.
- NO FOOD, DRINK, IN ANY FORM is allowed in the lab.
- TURN OFF CELL PHONES! If you need to use it, please keep it in bags.
- Avoid all horseplay in the laboratory. Do not misbehave in the computer laboratory. Work quietly.
- Save often and keep your files organized
- Do not reconfigure the cabling/equipment without prior permission.
- Do not play games on systems.
- If you finish early, spend the remaining time to complete the laboratory report writing. Come equipped with calculator and other materials related to lab works.
- Handle instruments with care. Report any breakage or faulty equipment to the Instructor. Shutdown your computer you have used for the purpose of your experiment before leaving the Laboratory.
- Violation of the above rules and etiquette guidelines will result in disciplinary action.

Varendra University

COURSE SYLLABUS

1	Faculty	Faculty of Science & Engineering
2	Department	Department of CSE
3	Program	B.Sc. in Computer Science and Engineering
4	Name of Course	Computer Architecture Lab
5	Course Code	CSE-324
6	Trimester and Year	Summer-2019
7	Pre-requisites	CSE-231
8	Status	Advanced Course
9	Credit Hours	1.5
10	Section	A+B+C
11	Class Hours	3 hours
12	Class Location	Room-403, Annex Building, Talaimari
13	Name (s) of Academic staff / Instructor(s)	**Umme Rumman Chaity & Ananna Hoque Shathi** Lecturer, Dept. of CSE, VU
14	Contact	chaiti@vu.edu.bd, sathi@vu.edu.bd
15	Counseling Hours	
16	Text Book	Carl Hamacher, Zvonko Vranesic and Safwat Zaky : **Computer Organization,** *McGraw-Hill*
17	Reference	*1.* John P. Hayes: **Computer Architecture and Organization,** *McGraw-Hill.* 2. William Stallings : **Computer Organization and Architecture: Designing for Performance,** *Prentice Hall*
18	Equipment & Aids	• Lab Sheet • Text Book
19	Course Description	This is an introductory course in Computer Architecture designed for students to become familiar with the fundamentals of computer architecture and their application to computer engineering problems. It provides essential tools that are needed from engineering professionals to measure a simple computer performance. It will cover basic concept of hardware organization, microinstructions and programming, memory and I/O organization, pipelining, components of central processing unit, interfacing with peripherals and composite architectures.
20	Course Objectives	• To cover basic concept of hardware organization. • To demonstrate various shifting operation using shift registers. • To teach read write operation of memory.

21	Learning Outcomes	1. Construct different types of counter using flip-flops. 2. Demonstrate data shifting operations using appropriate shift register. 3. Perform the experiment memory addressing and read/write operations.
22	Teaching Methods	Lecture, Video Demonstration, Problem Solving, Project Development
23	Topic Outline	

Class	Topics Or Assignments	CLOs	Reading Reference	Activities
Lab-1	An asynchronous counter using 74LS73 JK flip flop to count decimal 0-7.	1		Design, implement and discussion
Lab-2	An asynchronous counter using 74LS73 JK flip flop to count decimal 0-15.	1		Design, implement and discussion
Lab-3	A synchronous counter using 74LS73 JK flip flop to count decimal 0-2-4-6.	1		Design, implement and discussion
Lab-4	A synchronous counter using 74LS73 JK flip flop to count decimal 1-2-5-7.	1		Design, implement and discussion
Lab-5	serial to parallel shifting operation of 8-bit data using 8-bit shift register (74LS164).	2		Design, implement and discussion
Lab-6	parallel to serial shifting operation of 8-bit data using 8-bit shift register (74LS165).	2		Design, implement and discussion
Lab-7	Memory read write operation using IDT 6116 (Static RAM).	3		Design, implement and discussion

	have to follow the instructions given by the instructor for makeup. In case of absence in the mid/final exam for medical grounds, the student must also get his/her application forwarded by the head of the department before a make-up exam can be taken. It is recommended that the students inform the instructor beforehand through mail if they feel that they will miss a class/evaluation due to medical reasons.

<table>
<tr><td>25</td><td>Assessment Methods</td><td>
<table>
<tr><th>Assessment Types</th><th>Marks</th></tr>
<tr><td>Attendance</td><td>10%</td></tr>
<tr><td>Final Quiz</td><td>20%</td></tr>
<tr><td>Lab Report</td><td>10%</td></tr>
<tr><td>Final Lab Test</td><td>30%</td></tr>
<tr><td>Lab Viva-voce</td><td>10%</td></tr>
<tr><td>Regular Assessment and Performance</td><td>20%</td></tr>
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<tr><td>25</td><td>Grading Policy</td><td>
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<tr><th>Numerical Grade</th><th>Letter Grade</th><th>Grade Point</th></tr>
<tr><td>80% and above</td><td>A+</td><td>4.00</td></tr>
<tr><td>75% to less than 80%</td><td>A</td><td>3.75</td></tr>
<tr><td>70% to less than 75%</td><td>A-</td><td>3.50</td></tr>
<tr><td>65% to less than 70%</td><td>B+</td><td>3.25</td></tr>
<tr><td>60% to less than 65%</td><td>B</td><td>3.00</td></tr>
<tr><td>55% to less than 60%</td><td>B-</td><td>2.75</td></tr>
<tr><td>50% to less than 55%</td><td>C+</td><td>2.50</td></tr>
<tr><td>45% to less than 50%</td><td>C</td><td>2.25</td></tr>
<tr><td>40% to less than 45%</td><td>D</td><td>2.00</td></tr>
<tr><td>less than 40%</td><td>F</td><td>0.00</td></tr>
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</td></tr>
<tr><td>26</td><td>Additional Course Policies</td><td>1. 1. Lab Reports
Report on previous Experiment must be submitted before the beginning of new experiment. A bonus may be obtained if a student submits a neat, clean and complete lab report.
2. 2. Examination
There will be a lab exam at the end of the semester that will be closed book.
3. 3. Unfair means policy
In case of copying/plagiarism in any of the assessments, the students involved will receive zero marks. Zero Tolerance will be shown in this regard. In case of severe offences, actions will be taken as per university rule.
4. 4. Counseling
Students are expected to follow the counseling hours posted. In case of emergency/unavoidable situations, students can e-mail me to make an appointment. Students are regularly advised to check the piazza course page for updates/materials.
5. 5. Policy for Absence in Class/Exam
If a student is absent in the class for anything other than medical reasons, he/she will not receive attendance. If a student misses a class for genuine medical reasons, he/she must apply with the supporting documents (prescription/medical report). He/she will then</td></tr>
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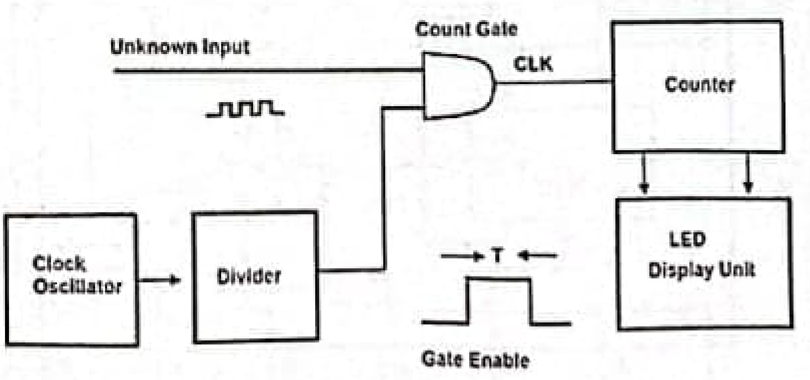
Lab No 1:

Counter Implementation Using Flip-flops

Counter and Flip-flops:

A counter is a device in computing and digital logic that is used to store and display the particular event so many times. The most common type of a counter is a sequential digital logic circuit. This circuit consists of one i/p line, namely clock and number of o/p lines. The values of the o/p lines denote a number in the BCD or binary number system. Generally, these circuits are designed with flip-flops which are connected in cascade. These devices are widely used in digital circuits and are designed as separate ICs and also combined as parts larger integrated circuits.



Types of Electronic Counters

Electronic counters can be implemented by using register type circuits like flip-flops and these are classified into different types and a few of them are discussed below.

- Synchronous counter
- Asynchronous Counter or Ripple Counter
- Ring counter
- Cascaded counter
- Johnson counters etc.

Asynchronous (Ripple) Counter

An asynchronous or ripple counter is used to store one bit, and counts from 0-1 before it overflows. Whenever the counter increases for every CLK cycle and takes two CLK cycles to overflow. So every cycle will change a transition from 0-1 and 1-0.

A 3-bit asynchronous counter is implemented using j-k flip-flops.

Task 1: Design and experiment an asynchronous counter using 74LS73 JK flip flop to count decimal 0-7.

Data Table:

Clock Pulse	J_{0-2}	K_{0-2}	Q2	Q1	Q0
1	High	High	0	0	0
2	High	High	0	0	1
3	High	High	0	1	0
4	High	High	0	1	1
5	High	High	1	0	0
6	High	High	1	0	1
7	High	High	1	1	0
8	High	High	1	1	1

*High=5 volts (1), Low=0 volts(0)

Lab No 2:

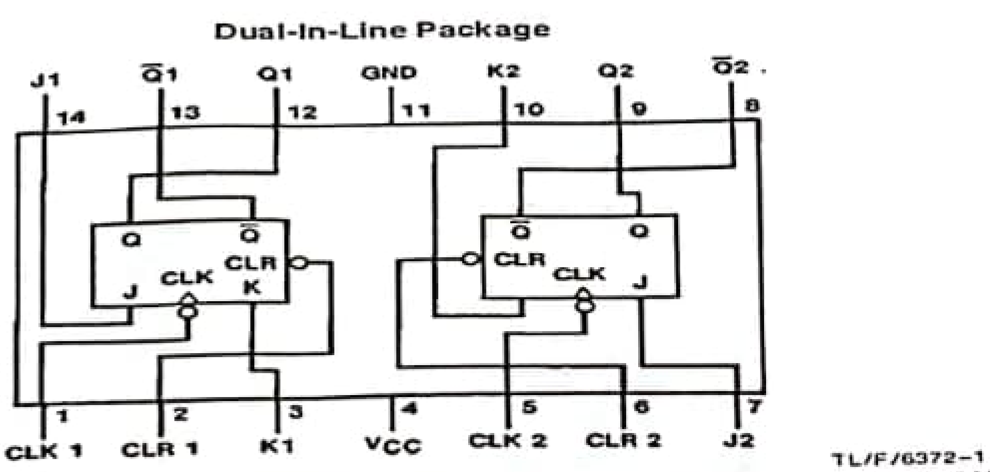
Counter Implementation Using Flip-flops

Task 1: Design and experiment an asynchronous counter using 74LS73 JK flip flop to count decimal 0-15.

Requirements:

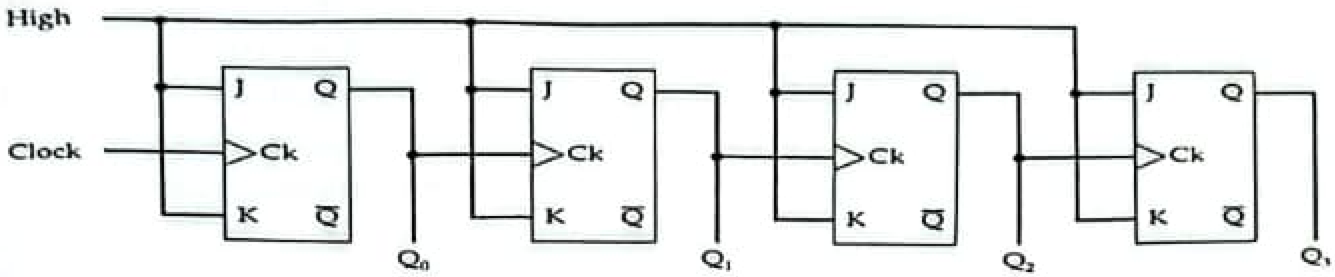
i) Trainer board
ii) IC(74ls73)- 2 pcs
iii) Wires

Pin Configuration:

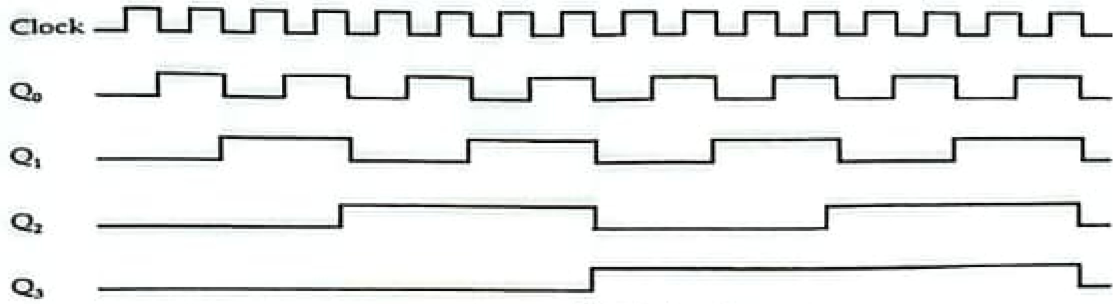


Order Number DM54LS73AJ, DM54LS73AW, DM74LS73AM or DM74LS73AN
See NS Package Number J14A, M14A, N14A or W14B

Circuit Diagram:



(a) Sequential Circuit



(b) Timing Diagram

Data Table:

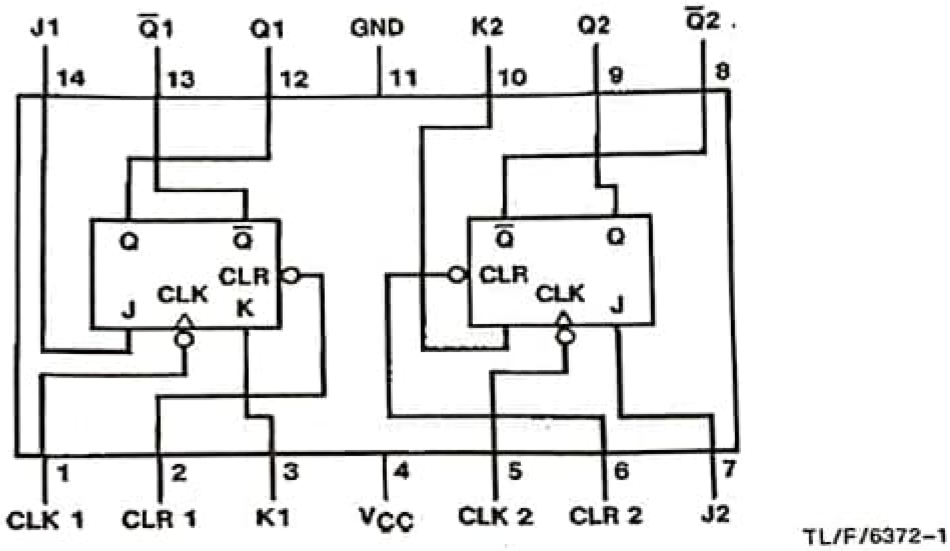
Clock Pulse	J_{0-2}	K_{0-2}	Q3	Q2	Q1	Q0
1	High	High	0	0	0	0
2	High	High	0	0	0	1
3	High	High	0	0	1	0
4	High	High	0	0	1	1
5	High	High	0	1	0	0
6	High	High	0	1	0	1
7	High	High	0	1	1	0
8	High	High	0	1	1	1
9	High	High	1	0	0	0
10	High	High	1	0	0	1
11	High	High	1	0	1	0
12	High	High	1	0	1	1
13	High	High	1	1	0	0
14	High	High	1	1	0	1
15	High	High	1	1	1	0
16	High	High	1	1	1	1

Requirements:

i) Trainer board
ii) IC(74ls73)- 2 pcs
iii) Wires

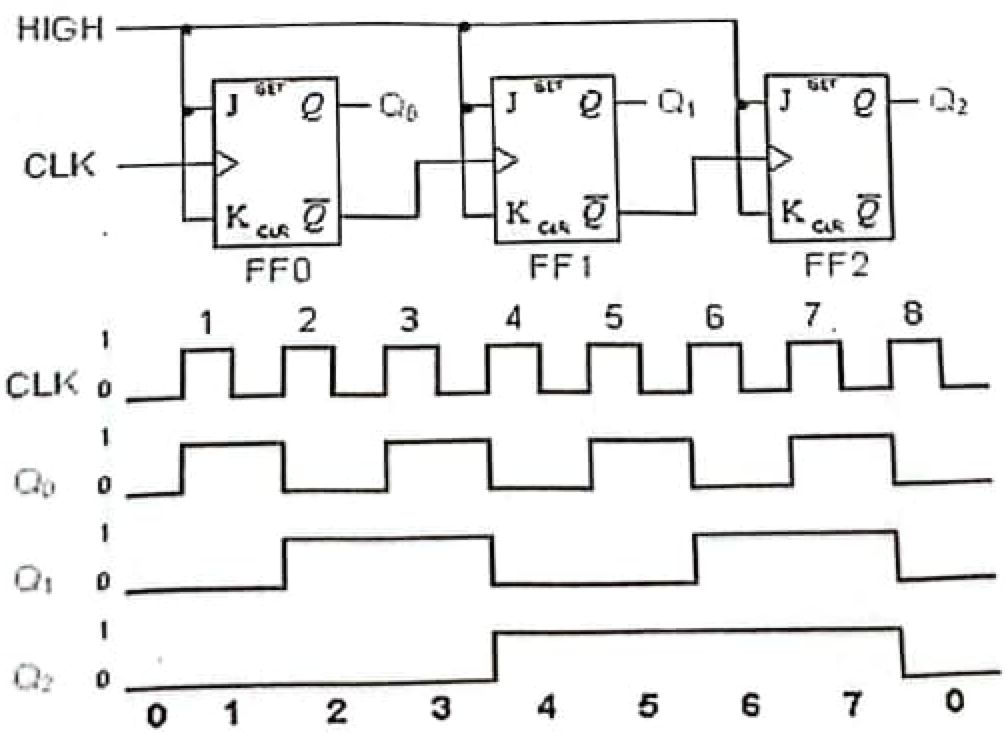
Pin Configuration:

Dual-In-Line Package



Order Number DM54LS73AJ, DM54LS73AW, DM74LS73AM or DM74LS73AN
See NS Package Number J14A, M14A, N14A or W14B

Circuit Diagram:



Lab No. 3:

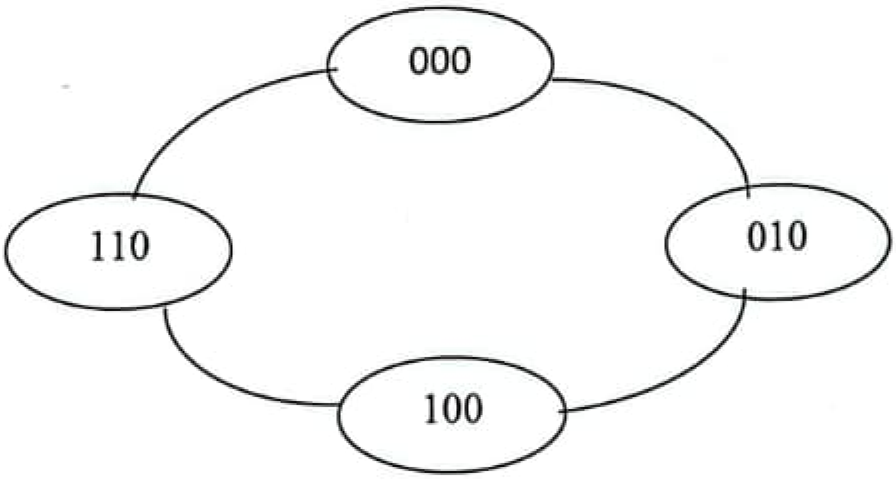
Synchronous Counter:

Synchronous generally refers to something which is coordinated with others based on time. Synchronous signals occur at same clock rate and all the clocks follow the same reference clock.

In previous lab experiment of Asynchronous Counter, we have seen that the output of that counter is directly connected to the input of next subsequent counter and making a chain system, and due to this chain system propagation delay appears during counting stage and create counting delays. **In synchronous counter,** the clock input across all the flip-flops use the same source and create the same clock signal at the same time. So, a counter which is using the same clock signal from the same source at the same time is called **Synchronous counter.**

Task 1: Design a 3 bit synchronous counter using 74LS73 JK flip flop to count decimal 0-2-4-6.

Step 1: State Diagram



Step 2: Next State Table-

Present State			Nest State		
Q2	Q1	Q0	Q2	Q1	Q0
0	0	0	0	1	0
0	1	0	1	0	0
1	0	0	1	1	0
1	1	0	0	0	0

Step 3: Flip –Flop Transition Table

Output Transition		Flip-Flop Input	
Q_N	Q_{N+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step 4: K-Map

Present			Next			Flip-Flop Input					
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	J_2	K_2	J_1	K_1	J_0	K_0
0	0	0	0	1	0	0	X	1	X	0	X
0	1	0	1	0	0	1	X	X	1	0	X
1	0	0	1	1	0	X	0	1	X	0	X
1	1	0	0	0	0	X	1	X	1	0	X

	$\overline{Q_0}$	Q_0
$\overline{Q_2Q_1}$	0	X
$\overline{Q_2}Q_1$	1	X
Q_2Q_1	X	X
$Q_2\overline{Q_1}$	X	X

$J_2 = Q_1$

	$\overline{Q_0}$	Q_0
$\overline{Q_2Q_1}$	X	X
$\overline{Q_2}Q_1$	X	X
Q_2Q_1	1	X
$Q_2\overline{Q_1}$	0	X

$K_2 = Q_1$

Task 2: Experiment a 3 bit synchronous counter using 74LS73 JK flip flop to count decimal 0-2-4-6.

Requirements:

- IC (74ls73)- 2 pcs
- Trainer board
- Connecting wires

Data Table:

Clock Pulse	J_0	K_0	J_1	K_1	J_2	K_2	Q2	Q1	Q0
1	0	1	1	1	Q_1	Q_1	0	0	0
2	0	1	1	1	Q_1	Q_1	0	0	1
3	0	1	1	1	Q_1	Q_1	0	1	0
4	0	1	1	1	Q_1	Q_1	0	1	1
5	0	1	1	1	Q_1	Q_1	1	0	0
6	0	1	1	1	Q_1	Q_1	1	0	1
7	0	1	1	1	Q_1	Q_1	1	1	0
8	0	1	1	1	Q_1	Q_1	1	1	1

Lab 4:

Design and experiment a synchronous counter using 74LS73 JK flip flop to count decimal 1-2-5-7.

Task 1: Home Assignment

Complete the steps of previous experiment to count 1-2-5-7.

	$\overline{Q_0}$	Q_0
$\overline{Q_2 Q_1}$	X	X
$\overline{Q_2} Q_1$	1	X
$Q_2 Q_1$	1	X
$Q_2 \overline{Q_1}$	X	X

$K_1 = 1$

	$\overline{Q_0}$	Q_0
$\overline{Q_2 Q_1}$	1	X
$\overline{Q_2} Q_1$	X	X
$Q_2 Q_1$	X	X
$Q_2 \overline{Q_1}$	1	X

$J_1 = 1$

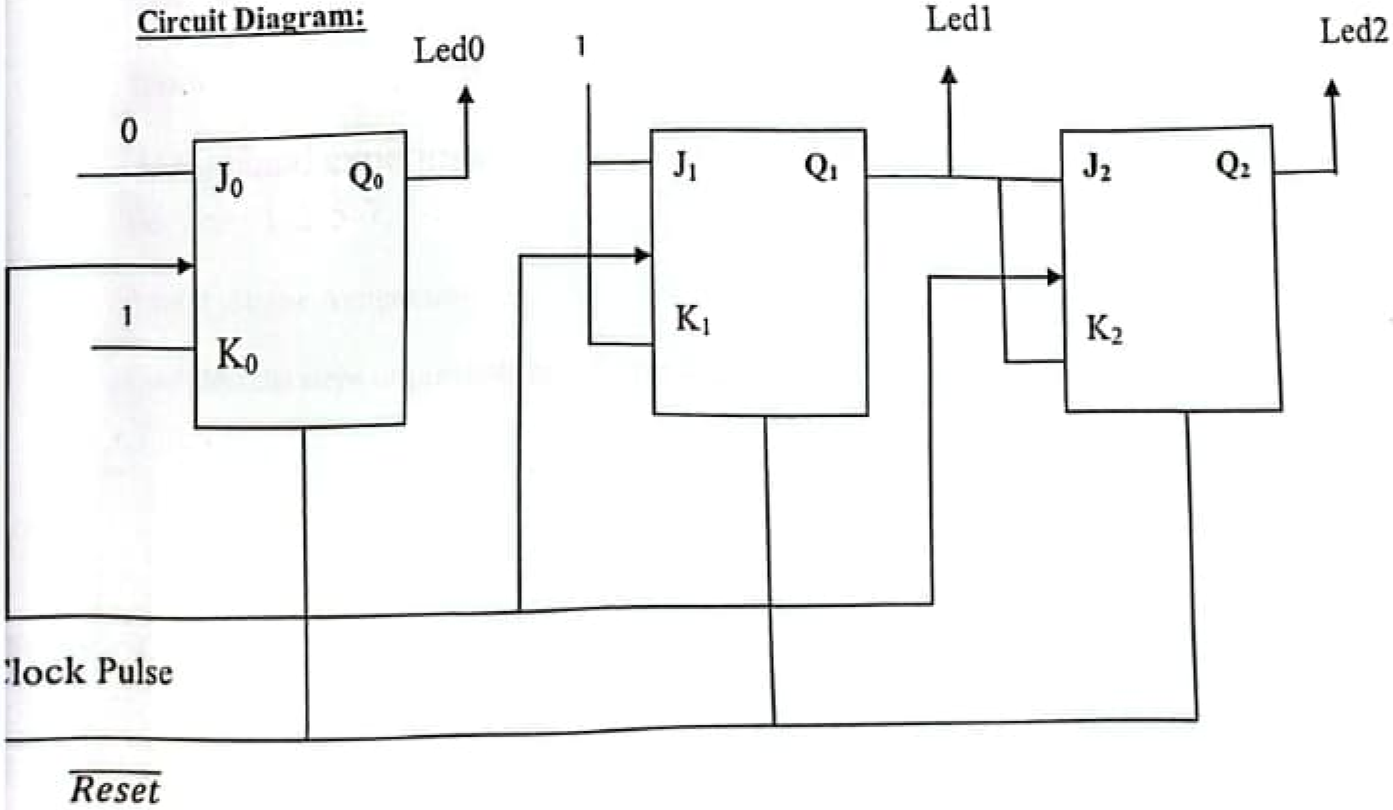
	$\overline{Q_0}$	Q_0
$\overline{Q_2 Q_1}$	0	X
$\overline{Q_2} Q_1$	0	X
$Q_2 Q_1$	0	X
$Q_2 \overline{Q_1}$	0	X

$J_0 = 0$

	$\overline{Q_0}$	Q_0
$\overline{Q_2 Q_1}$	X	X
$\overline{Q_2} Q_1$	X	X
$Q_2 Q_1$	X	X
$Q_2 \overline{Q_1}$	X	X

$K_0 = 0/1$

Circuit Diagram:

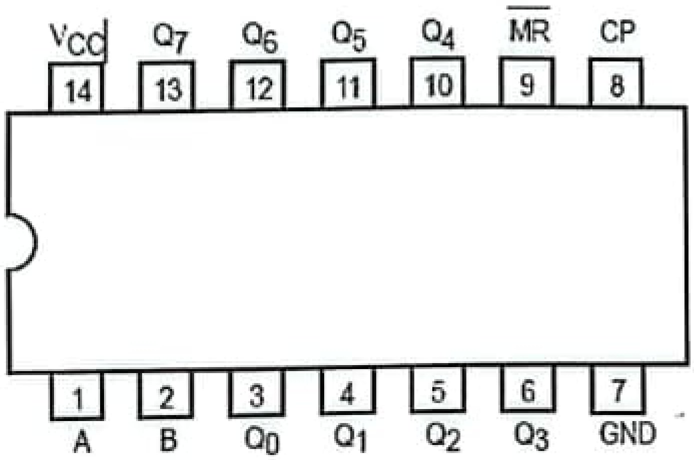


Task 1:

Design and experiment serial to parallel shifting operation of 8-bit data using 8-bit shift register (74LS164).

Pin configuration:

CONNECTION DIAGRAM DIP (TOP VIEW)



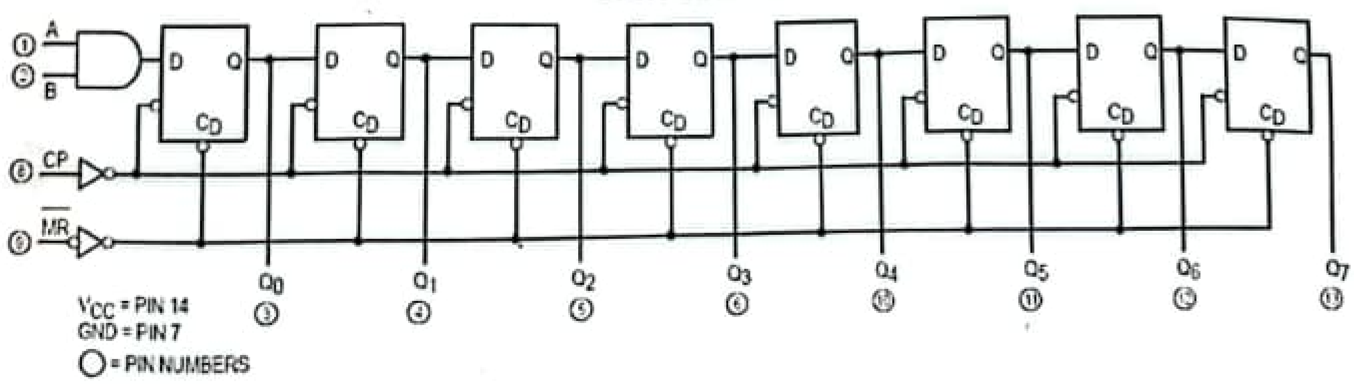
NOTE:
The Flatpak version has the same pinouts (Connection Diagram) as the Dual In-Line Package.

Requirements:

- IC (74ls164)- 1 pc
- Trainer board
- Connecting wires

Internal Circuit Diagram:

LOGIC DIAGRAM



Lab 5:

Shift Register and Their operations

The Shift Register is another type of sequential logic circuit that can be used for the storage or the transfer of binary data .This sequential device loads the data present on its inputs and then moves or "shifts" it to its output once every clock cycle, hence the name **Shift Register.**

Shift Registers are used for data storage or for the movement of data and are therefore commonly used inside calculators or computers to store data such as two binary numbers before they are added together, or to convert the data from either a serial to parallel or parallel to serial format. The individual data latches that make up a single shift register are all driven by a common clock (Clk) signal making them synchronous devices.Shift register IC's are generally provided with a *clear* or *reset* connection so that they can be "SET" or "RESET" as required. Generally, shift registers operate in one of four different modes with the basic movement of data through a shift register being:

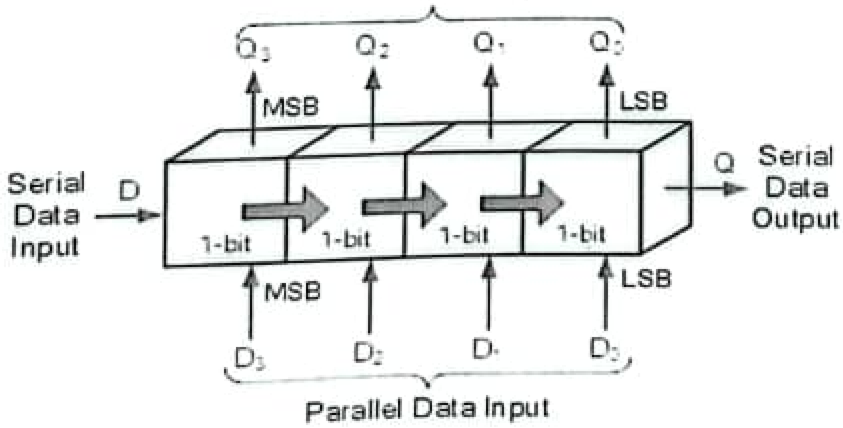
Serial-in to Parallel-out (SIPO) - the register is loaded with serial data, one bit at a time, with the stored data being available at the output in parallel form.

Serial-in to Serial-out (SISO) - the data is shifted serially "IN" and "OUT" of the register, one bit at a time in either a left or right direction under clock control.

Parallel-in to Serial-out (PISO) - the parallel data is loaded into the register simultaneously and is shifted out of the register serially one bit at a time under clock control.

Parallel-in to Parallel-out (PIPO) - the parallel data is loaded simultaneously into the register, and transferred together to their respective outputs by the same clock pulse.

The effect of data movement from left to right through a shift register can be presented graphically as:



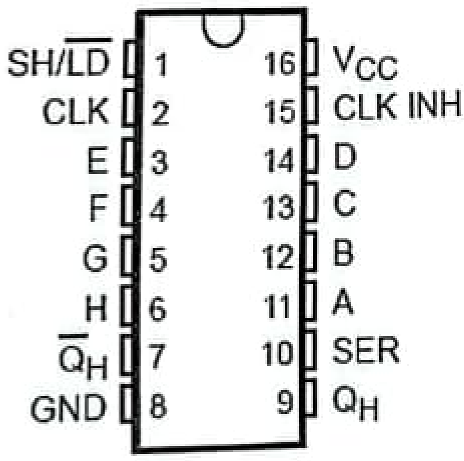
Data Table:

Clk	A	B	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7
0	1	1	1	0	0	0	0	0	0	0
1	1	1	1	1	0	0	0	0	0	0
2	0	1	0	1	1	0	0	0	0	0
3	1	0	0	0	1	1	0	0	0	0
4	1	1	1	0	0	1	1	0	0	0
5	0	0	0	1	0	0	1	1	0	0
6	1	1	1	0	1	0	0	1	1	0
7	0	1	0	1	0	1	0	0	1	1

Lab 6:

Design and experiment parallel to serial shifting operation of 8-bit data using 8-bit shift register (74LS165).

Pin Configuration:



Requirements:

- IC (74ls165)- 1 pc
- Trainer board
- Connecting wires

Data Table:

Clk	CLKINH	SH/$\overline{LD}$	A	B	C	D	E	F	G	H	Q_H
0	1	0	1	0	1	0	1	1	0	0	
1	0	1	1	0	1	0	1	1	0	0	0
2	0	1	1	0	1	0	1	1	0	0	0
3	0	1	1	0	1	0	1	1	0	0	1
4	0	1	1	0	1	0	1	1	0	0	1
5	0	1	1	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	1	1	0	0	1
7	0	1	1	0	1	0	1	1	0	0	0
8	0	1	1	0	1	0	1	1	0	0	1

Lab 7:

Memory Read Write Operation

A memory unit stores binary information in groups of bits called words. Data input lines provide the information to be stored into the memory, Data output lines carry the information out from the memory. The control lines Read and write specifies the direction of transfer of data.

Basically, in the memory organization, there are memory locations indexing from 0 to 2^l -1 Where l is the address buses. We can describe the memory in terms of the bytes using the following formula:

$N=2^l$ Bytes
Where,
l is the total address buses
N is the memory in bytes

Memory Read Operation:
Memory read operation transfers the desired word to address lines and activates the read control line.

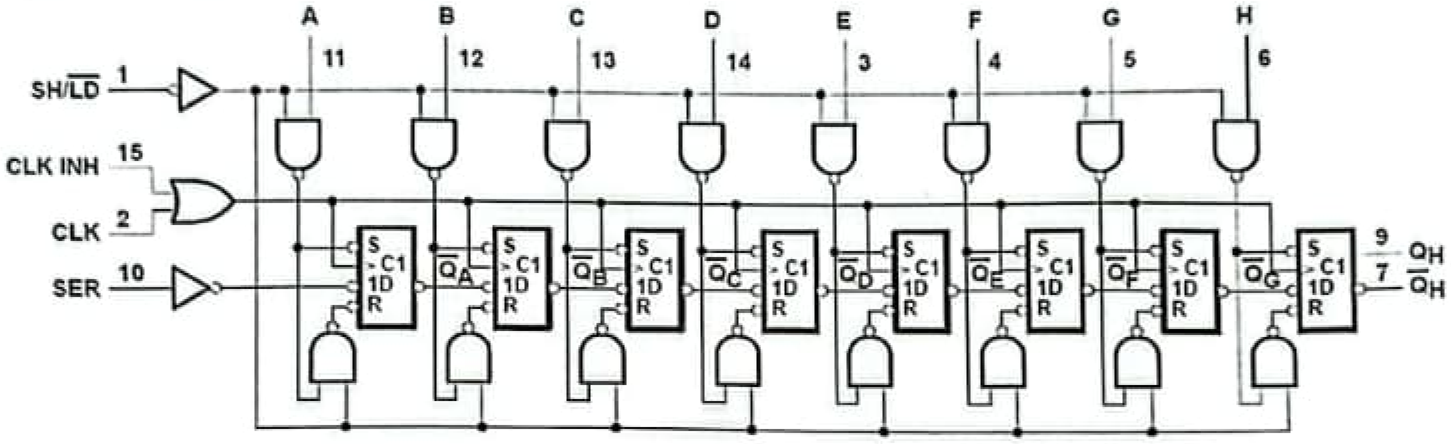
Memory Write Operation:
Memory write operation transfers the address of the desired word to the address lines, transfers the data bits to be stored in memory to the data input lines. Then it activates the write control line.

Task 1:

Design and Experiment Memory read write operation using IDT 6116 (Static RAM).

Internal Circuit Diagram:

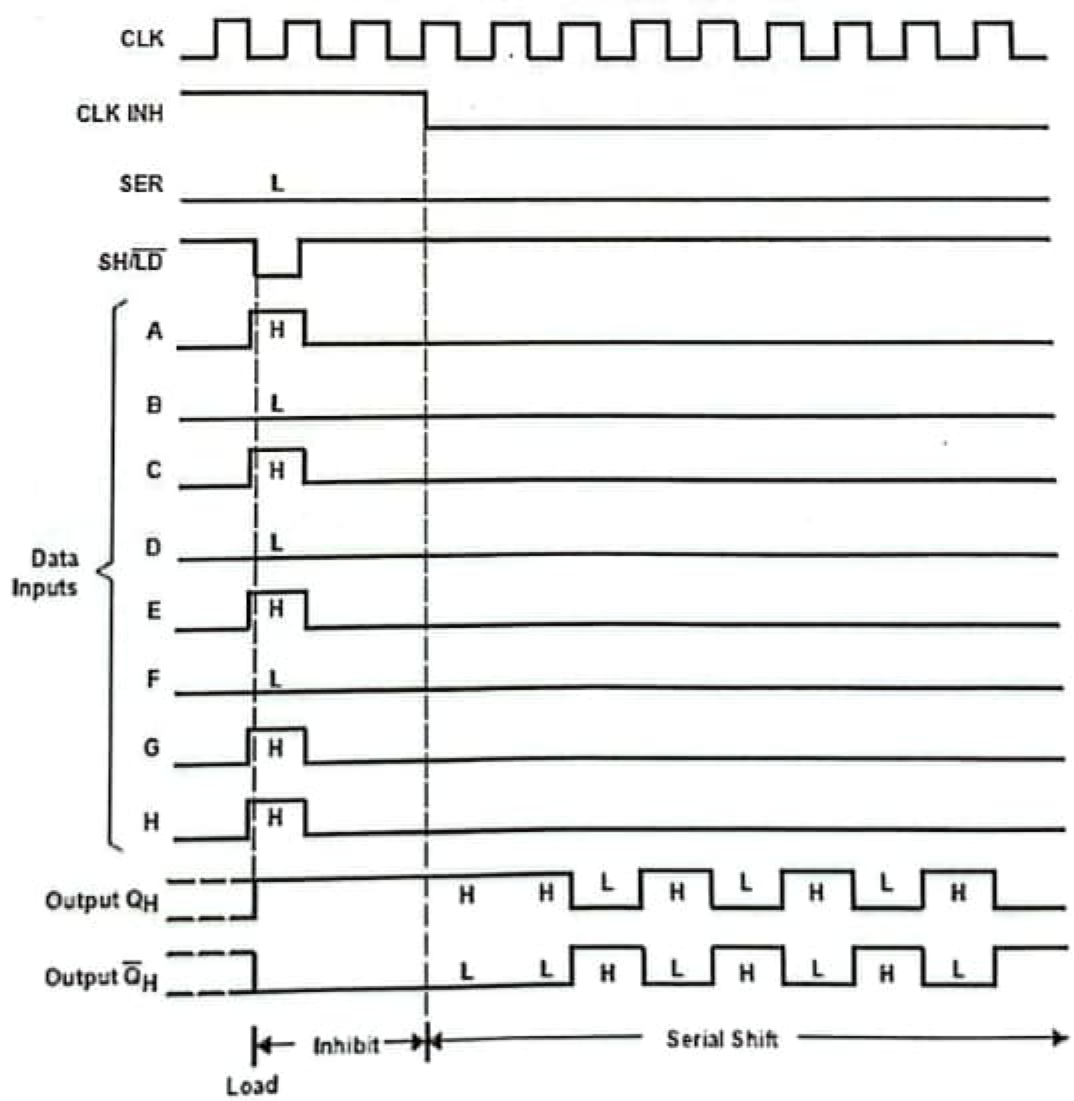
logic diagram (positive logic)



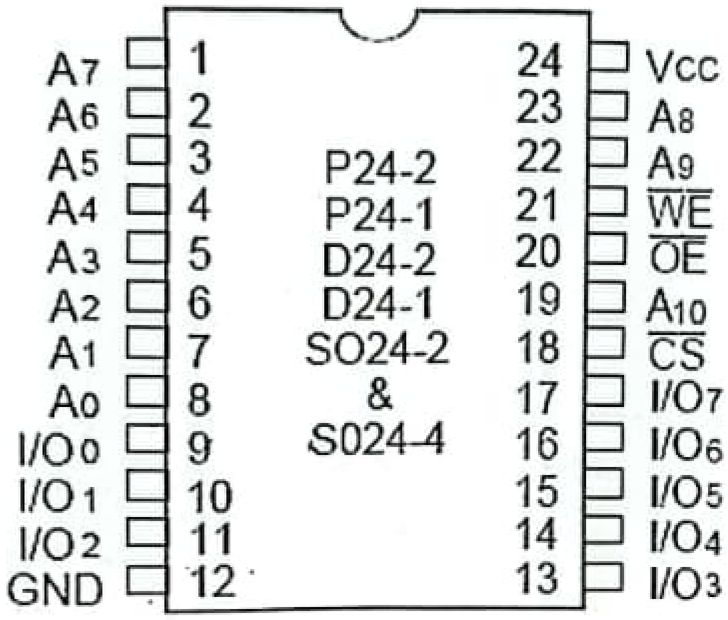
Pin numbers shown are for D, J, N, NS, and W packages.

Timing Diagram:

typical shift, load, and inhibit sequences



Pin Configuration:



Requirements:

- IC (IDT 6116)- 1 pc
- Trainer board
- Connecting wires

Data Table: Write Data

$\overline{CS}$	$\overline{WE}$	$\overline{OE}$	$A_{10} - A_2$	A_1	A_0	$D_7 - D_4$	D_3	D_2	D_1	D_0
0	0	1	0	0	0	0	0	0	0	1
0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	1	0	0	0	1	1	1
0	0	1	0	1	1	0	1	1	1	1

Data Table: Read Data

$\overline{CS}$	$\overline{WE}$	$\overline{OE}$	$A_{10} - A_2$	A_1	A_0	$D_7 - D_4$	D_3	D_2	D_1	D_0
0	1	0	0	0	0	0	0	0	0	1
0	1	0	0	0	1	0	0	0	1	1
0	1	0	0	1	0	0	0	1	1	1
0	1	0	0	1	1	0	1	1	1	1